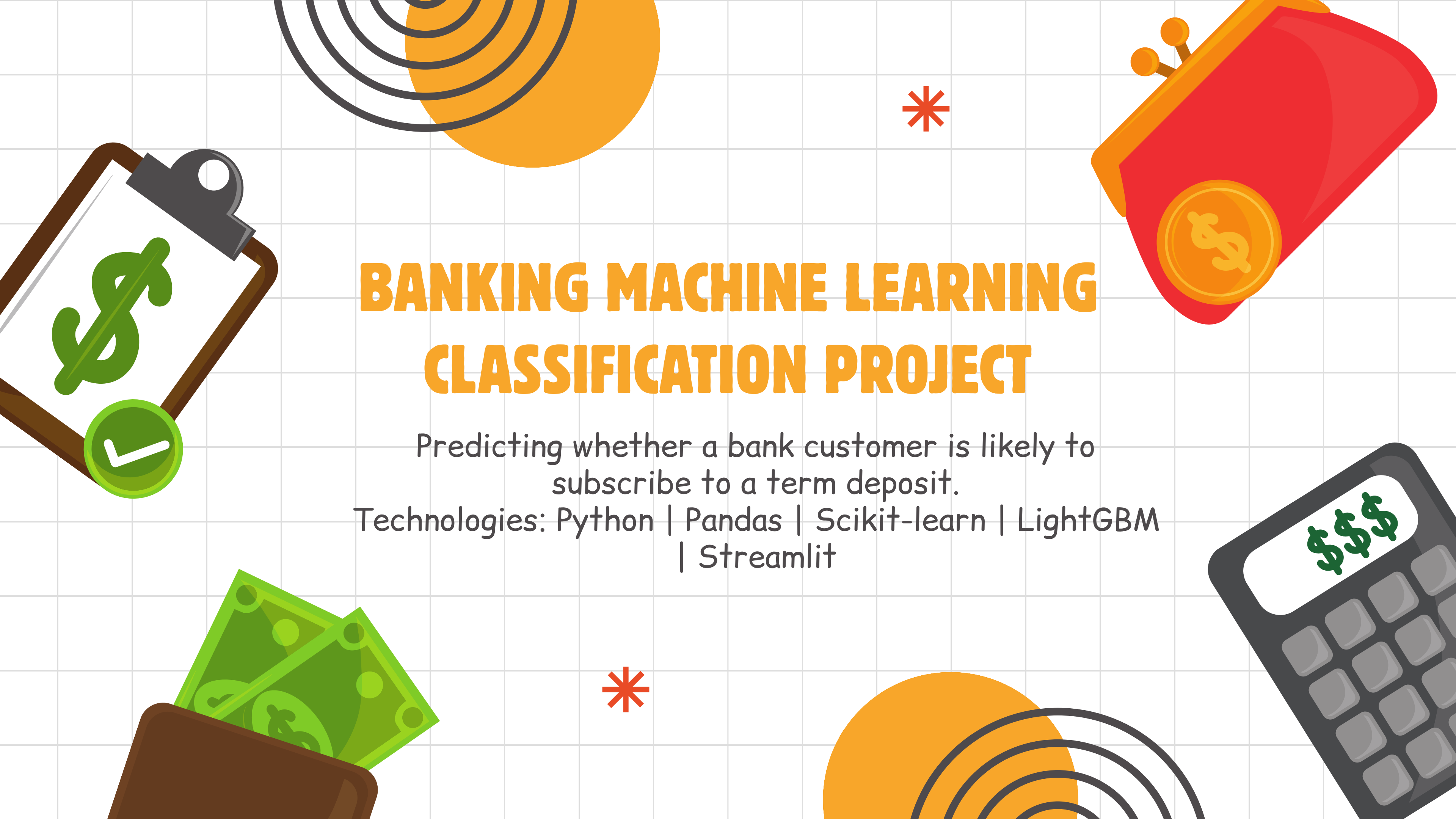




BANKING MACHINE LEARNING CLASSIFICATION PROJECT

Predicting whether a bank customer is likely to
subscribe to a term deposit.

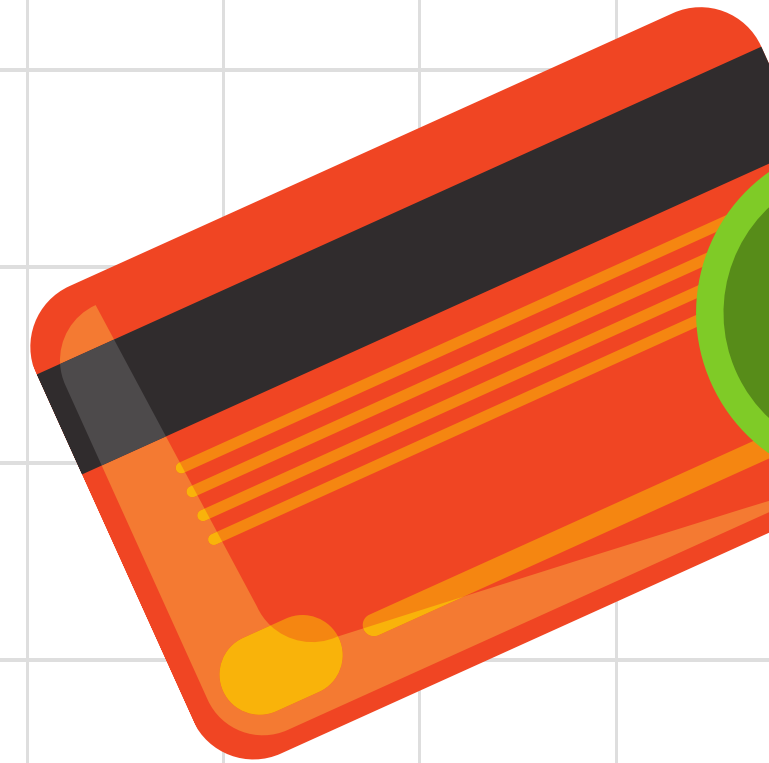
Technologies: Python | Pandas | Scikit-learn | LightGBM
| Streamlit



Introduction & Problem Statement



Context: Banks conduct direct marketing campaigns to promote term deposits, but contacting all customers equally is inefficient.



The Problem:

- High marketing costs & wasted resources.
- Low conversion rates and customer annoyance from
unnneeded calls.
- Core Question: How can we predict customer
subscription likelihood before or during a marketing
campaign?





Project Objectives *

- Develop an end-to-end Machine Learning pipeline to predict term deposit subscription.
- Clean, preprocess, and handle complex real-world data distributions.
- Train, compare, and evaluate multiple classification algorithms.
- Select the best-performing model and deploy it via an interactive Streamlit web app.



Dataset Overview & Target



- Dataset: Bank marketing data covering demographics, financial metrics, and campaign details.
- Key Features:
- Demographics: Age, Job, Marital Status, Education.
- Financials: Bank Balance, Housing Loan, Personal Loan, Default.
- Campaign: Contact Method, Month, Day, Duration, Previous Contacts, Poutcome.
- Target Variable (Deposit): Binary (0 = Will Not Subscribe, 1 = Will Subscribe).





Data Preprocessing (Cleaning & Encoding)



Missing & Unknown Values:

- Categorical: Imputed using Most Frequent (Mode).
- Numerical: Imputed using Mean (normal distribution) or Median (skewed distribution).

-

Categorical Encoding:

- Label Encoding: Binary fields (housing, loan, etc.).
- Ordinal Encoding: Ordered variables (education, month).
- One-Hot Encoding: Nominal variables (marital).
- Target Encoding: High-cardinality categories (job).



Outlier Treatment & Feature Scaling



Outlier Handling:

- Analyzed via Box Plots.
- Applied IQR Capping (capping extreme values at calculated boundaries) to preserve data without dropping records.

-

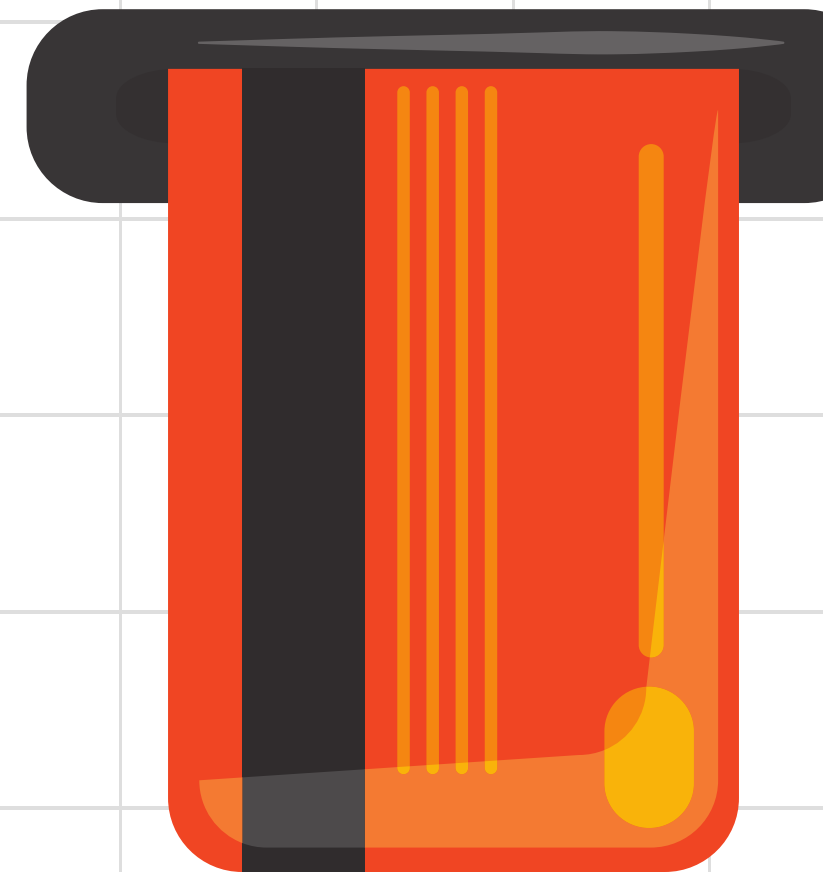
Feature Scaling:

- Applied tailored scaling (StandardScaler, RobustScaler, or MinMaxScaler) based on feature distribution to optimize algorithm performance.



Feature Selection

- SelectPercentile (ANOVA F-test): Identified statistically significant features correlated with the target variable (f_classif).
- Random Forest Feature Importance: Measured tree-based feature importance to retain the most predictive demographic and campaign attributes.





Machine Learning Models Tested



A wide range of baseline, tree-based, and boosting algorithms were evaluated:

- Linear & Distance Models: Logistic Regression, KNN, SVM, Naive Bayes.
- Tree & Ensemble Models: Decision Tree, Random Forest, Extra Trees, AdaBoost, Gradient Boosting, HistGradientBoosting, XGBoost, CatBoost, LightGBM.





Model Evaluation Metrics



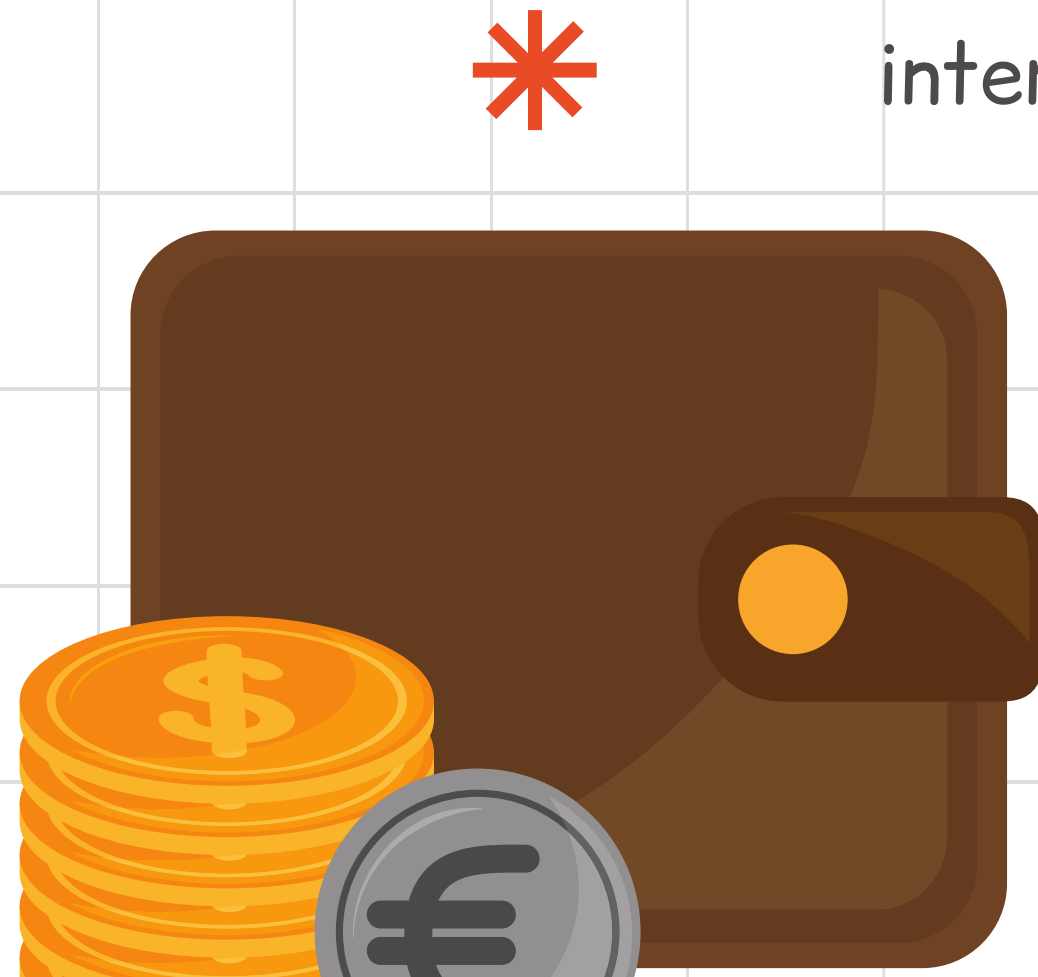
Models were benchmarked using comprehensive classification metrics:

- Accuracy & Balanced Accuracy (Handling class balance)
- Precision & Recall (Minimizing false positives vs. capturing true subscribers)
- F1-Score (Balance between Precision and Recall)
- ROC-AUC (Model discrimination capability)



Model Comparison & Selection

- All 13 models were systematically evaluated across all metrics. *
- Final Choice: LightGBM
- Why LightGBM?
 - Achieved the highest overall performance (F1-Score / ROC-AUC).
 - Exceptional speed and efficiency on tabular data.
 - Robust handling of non-linear feature interactions.

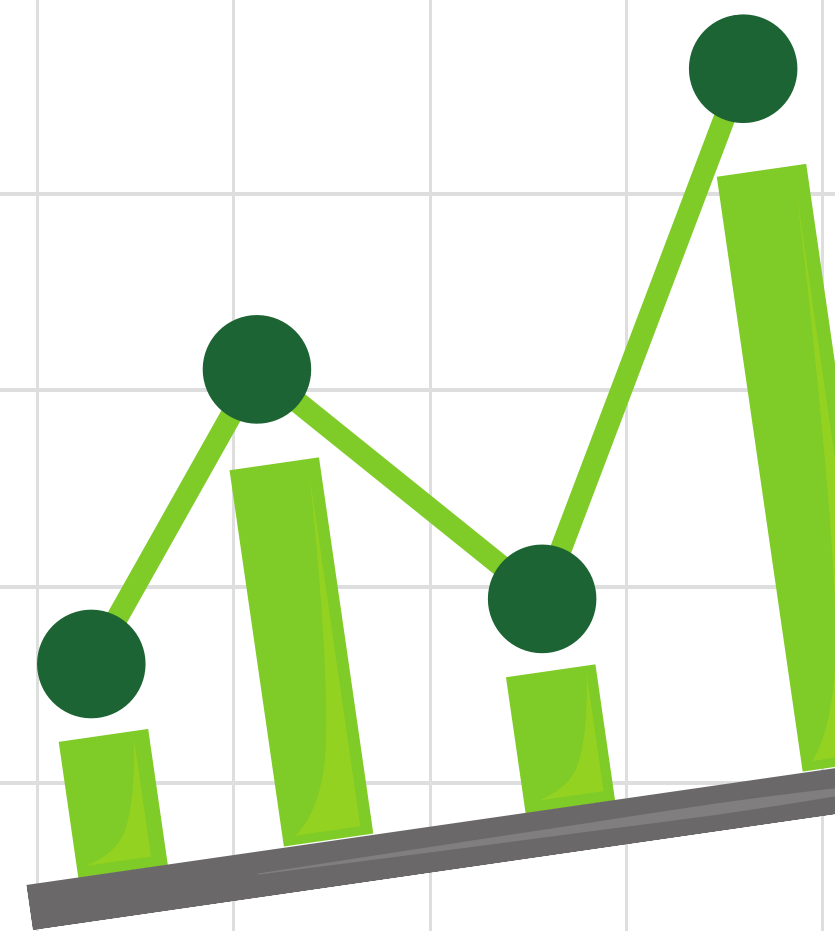


End-to-End ML Pipeline



The system uses an integrated pipeline to ensure reproducible data transformation:

Raw Customer Data → Imputation → Encoding



Streamlit Application *



- Interactive UI: Built using Streamlit for instant predictions.
- Input Fields: Users input customer demographic, financial, and contact campaign details.
- Output Provided:
 - a. Binary Prediction: (Will Subscribe / Will Not Subscribe).
 - b. Probability Score (%): Displays the confidence score of the model.



Business Impact

- Targeted Marketing: Focus sales efforts on high-probability leads.
- Cost Reduction: Drastically decrease contact/call center operational costs.
- Higher Conversion Rates: Maximize ROI by reaching receptive customers.
- Resource Optimization: Efficient allocation of sales team time.



Limitations & Future Work



- Limitations: Performance relies on data quality; campaign contact features are collected mid-process.
- Future Work:
 - Integrate Explainable AI (SHAP) for model interpretability.
 - Implement Batch Predictions (CSV file uploads).
 - Deploy online with automated periodic retraining.





CONCLUSION & THANK YOU

- Successfully built an end-to-end ML solution from data cleaning to model deployment.
- LightGBM delivered superior predictive performance and seamlessly integrated into Streamlit.

Thank You!
Questions & Discussion